Yanyu Wang

Ph.D. Candidate
University of Maryland Center for Environmental Science
301 Braddock Rd, Frostburg, MD 21532

EDUCATION

University of Maryland Center for Environmental Science *July 2021 to Present*
Ph.D. in Marine, Estuarine, and Environmental Sciences (MEES)
- Project title: "INFEWS: U.S.-China: Managing agricultural nitrogen to achieve sustainable Food-Energy-Water Nexus in China and the U.S."
- Advisor: Dr. Xin Zhang

Emory University- Atlanta, GA *August 2019 to May 2021*
M.S. in Environmental Sciences
- Thesis title: "Agricultural Greenhouse Gas Fluxes Under Different Cover Crop Systems"
- Advisor: Dr. Eri Saikawa

Colorado State University - Fort Collins, CO *August 2017 to May 2019*
B.S. in Ecosystem Science and Sustainability

Anhui Agriculture University - Hefei, China *September 2015 to June 2017*
B.S. in Environmental Science

HONORS AND GRANTS

Ann G. Wylie Dissertation Fellowship ($15,000) – University of Maryland *Spring 2025*
- The Fellowship supports University of Maryland doctoral candidates who have excellent qualifications and are in the latter stages of writing their dissertations.

Debbie Morrin-Nordlund Memorial Award ($1,600) – University of Maryland *Summer 2024*
- The grant is granted to support MEES students to present work at professional meetings or travel to conduct research.

Dean's Fellowship ($10,000) – University of Maryland *Summer 2024*
- This highly competitive and prestigious fellowship is awarded by the University of Maryland Graduate School to students based on their academic achievements and progress toward their graduate degree.

Lester Research Grants ($5,000) – Emory University *Summer 2020*
- The grant is granted to support summer fieldwork of measuring agricultural greenhouse gas emissions from cover crop systems at University of Georgia

Dean's List – Colorado State University *November and January 2018*

The Water Scholars Program Scholarship ($50,000) – Coca-Cola *August 2017*
- Received full tuition prestigious scholarship for academic excellence and promise in sustainability and water research

PROFESSIONAL EXPERIENCE

Summer intern – Jornada experimental range, NM *June to July 2025*
- Contributed to manure recycling research by assisting with manure application on farms and conducting greenhouse experiments
- Contributed to research on manure excretion and recycling estimates as part of the Nitrogen 2.0 project, funded by Spark Climate Solutions
- Advisor: Dr. Sheri Spiegal

Graduate Research Assistant – UMCES-Appalachian Lab *July 2021 to Present*

- Lead the research work for a NSF funded project: "INFEWS: U.S.-China: Managing agricultural nitrogen to achieve sustainable Food-Energy-Water Nexus in China and the U.S."
- Advisor: Dr. Xin Zhang

Teaching Assistant – Emory University *August 2020 to May 2021*

- Climate Change and Society (ENVS 326/526) – Dr. Eri Saikawa
- Fund. Concepts in Soil Sci (ENVS 285) – Dr. Debjani Sihi

Responsibilities: facilitate discussion in the break-out room; prepare answer keys; programming assignments, exams, or projects; grade assignments; hold weekly office hours

ESS Club International Coordinator – Colorado State University *Sep 2019 to May 2019*

- Served as a coordinator to engage some international students into some events of ESS club
- Promoted the relationship between local students and international students

OTHER RESEARCH EXPERIENCE

Early career/young scientist convener – American Geophysical Union 2024 *December 2024*

- Developed and executed a social media strategy to promote the session.
- Facilitated networking opportunities among session participants.

Stakeholder workshop organizer – Institute of Marine and Environmental Technology *November 2022*

- Identified and engaged potential participants with complementary expertise.
- Developed the workshop workflow and contributed to facilitating discussions.

Undergraduate researcher, USDA-ARS Fort Collins *Jan to Dec 2018*

- Tested user-friendly GHG prediction model (COMET-farm) and compared the outputs with measured data under the supervision of Dr. Steve Del Grosso and Dr. Catherine Stewart.

PUBLICATIONS

Wang, Y., X. Zhang, S. Spiegal, E. Davidson. Informing manure recycling with an integrated indicator framework. *Nature Food*. Under Revision.

Wang, Y., X. Zhang, E. Davidson, B. Gu. Shifting trade from feed to food reduces agricultural nitrogen loss and GHG emissions in U.S. and China. *Environmental Research Letters*. In Review.

Zhang, X, R.Sabo, L.Rosa, H.Niazi, P.Kyle, J.Byun, **Y.Wang**, X. Yan, B. Gu, E. Davidson. Nitrogen management during decarbonization. Nat Rev Earth Environ 5, 717–731 (2024). https://doi.org/10.1038/s43017-024-00586-2

Zhang, X., **Y.Wang**, L. Schulte-Uebbing, W. De Vries, T. Zou, and E. A. Davidson. Sustainable Nitrogen Management Index: Definition, Global Assessment and Potential Improvements. *Front. Agr. Sci. Eng.*, **9**(3): 356–365 https://doi.org/10.15302/J-FASE-2022458, 2022

Wang, Y., Saikawa. E, Avramov. A, and Hill. NS. Agricultural Greenhouse Gas Fluxes Under Different Cover Crop Systems. *Front. Clim.*, **3**, 742320. doi: 10.3389/fclim.2021.742320, 2022

CONFERENCE PRESENTATIONS

Y.Wang. Nitrogen budget in the contiguous U.S. and its decadal shifts from 2012 to 2022. AGU 2024. Washington D.C., Dec, 2024

Y.Wang. Shifting trade from feed to food reduces agricultural nitrogen loss and GHG emissions in U.S. and China. University of Illinois. Champaign, July, 2024

Y.Wang. Zhang, X., Davidson, E. Reintegrating livestock and crop production to achieve sustainable Energy- Water-Land (EWL) nexus. AGU Fall meeting 2023. Oral Session, San Francisco, December, 2023

Y.Wang. Importing feed or food? The impacts of shifting trade portfolio between the US and China. The XXI International N Workshop. Madrid, Spain, Oct, 2022

Y.Wang. Different cover crop systems potential in mitigating agricultural GHG emissions. AGU Fall meeting 2020. Oral Session, Dec, 2020

Y.Wang. Trace gas fluxes under different cover crop systems. 7th Annual Southeastern Biogeochemistry Symposium. Mar, 2020

Y.Wang. Evaluation of the COMET-Farm decision support system and the DayCent ecosystem model using field observations. 24th Celebrate Undergraduate Research and Creativity. Apr, 2018